

main.py

Visualize

Run

```

1 #MOHAMMED SHAHID S
2 from queue import PriorityQueue
3
4
5 def best_first_Search(graph, start, target, heuristics):
6     visited = set()
7     pq = PriorityQueue()
8     # (heuristic_value, current_node, path)
9     pq.put((heuristics[start], start, [start]))
10    visited.add(start)
11
12    while not pq.empty():
13        h, node, path = pq.get()
14
15        if node == target:
16            return path, h
17
18        for neighbor in graph.get(node, []):
19            if neighbor not in visited:
20                visited.add(neighbor)
21                pq.put((heuristics[neighbor], neighbor, path + [neighbor]))
22
23    return None, None
24
25

```

Output

Path found by Best-First Search: ['A', 'C', 'F'] with target heuristic: 0

=== Code Execution Successful ===